THE CATEGORY OF DIVIDED POWER RINGS HAS ALL COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. 'F commutes with limits' is a one-line universal-property identity, not a gap. Declared dependency: lemma-a-version-of-brown is not carded (pure pointer to the categories.tex original); recorded via external_references.

Allowed standard prerequisites (editorial).

- Brown representability, divided power version. lemma-a-version-of-brown (same chapter, named; NOT carded – build-gate exclusion). Representation criterion applied to the functor F; declared as an external reference, not re-proved. The general categories.tex version is carded cross-chapter as S9-categories-a-version-of-brown.

Source reference. Original lemma 07GW: sources/stacks-dpa/pinned-original-source.tex, source lines 600–600. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The category of divided power rings has all colimits.*

Proof. The empty colimit is $\mathbf{Z}$ with divided power ideal (0). Let's discuss general colimits. Let $\mathcal{C}$ be a category and let $c \mapsto (A_c, I_c, \gamma_c)$ be a diagram. Consider the functor

$$F(B, J, \delta) = \lim_{c \in \mathcal{C}} Hom((A_c, I_c, \gamma_c), (B, J, \delta))$$

Note that any $f = (f_c)_{c \in C} \in F(B, J, \delta)$ has the property that all the images $f_c(A_c)$ generate a subring B' of B of bounded cardinality κ and that all the images $f_c(I_c)$ generate a divided power sub ideal J' of B'. And we get a factorization of f as a f' in $F(B')$ followed by the inclusion $B' \to B$. Also, F commutes with limits. Hence we may apply Lemma 07GW to see that F is representable and we win. □

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.